

CLOUD, INFRASTRUCTURE & CYBERSECURITY ENGINEERING

Unified professional track combining cloud architecture, system administration, cloud platform development, DevOps, data center operations, and cybersecurity into one integrated career path.

The unified track prepares graduates to design, implement, secure, automate, and operate modern enterprise and cloud platforms. It blends Linux and Windows administration, networking, virtualization, data center technologies, cloud services, DevOps and platform engineering, backend/cloud-native development, and defensive/offensive cybersecurity. The curriculum emphasizes practical labs, real-world projects, certification readiness, and AI-assisted operations for monitoring, troubleshooting, security analysis, and automation.

JOB PROFILES

Cloud Engineer · Cloud Administrator · Cloud Support Engineer · Cloud Platform Developer · DevOps Engineer · SRE / Platform Engineer · Data Center Engineer · System Administrator · Computer Networks Engineer · Database Administrator · Backend Developer · Cybersecurity Engineer · Cyber Threat Analyst · Web Application Penetration Tester · Mobile Application Penetration Tester · Incident Responder & Digital Forensics Analyst · GRC Specialist · DevSecOps Engineer

TOTAL HOURS	DELIVERY	UNIVERSITY GRADE	CERTIFICATION PATHS
1,800 hrs	Blended	Good	10+
Theory + hands-on labs + project	75% face-to-face 25% online	Minimum grade required	Cloud · Platform · Security

Where graduates work

WE · Vodafone Egypt · VOIS · Etisalat Misr · Orange · Fawry · Aman · PayMob · MaxAB · NBE · Bank Misr · Alex Bank · Suez Canal Bank · EJADA · DataScience · Huawei · Zinad · Fixed Solution · Inovasys · CyShield · Security Meter · EFG Holding · PwC · Deloitte · Advansys · Banque du Caire

Certification options

- RHCSA - Red Hat Certified System Administrator
- AWS Certified Cloud Practitioner
- AWS-SAA-C03 - AWS Certified Solutions Architect - Associate
- Microsoft Azure Administrator
- CKA - Certified Kubernetes Administrator
- NCP-MCI - Nutanix Certified Professional - Multicloud Infrastructure
- HCCDA - Huawei Certified Cloud Developer Associate

Selection process

01 IQ & English Exam - problem-solving aptitude and English proficiency.
 02 Technical Exam - core fundamentals: networking, Linux, cloud, DB, web, programming/OOP, and cybersecurity basics.
 03 Technical Interview - discussion of pre-work and track readiness.
 04 Interpersonal Interview - final communication and professional fit assessment.

Who may apply

Open to graduates of Computer Engineering, Communications Engineering, Computer Science, and relevant Engineering specializations with a minimum grade of Good or higher.

Program content - 1,800 hours

Hours are consolidated from the five source tracks after removing repeated topics and keeping advanced/specialized variants where they add new professional outcomes.

CATEGORY	HOURS	SHARE	KEY COURSES / DOMAINS
Fundamental Courses	240 hrs	13%	Operating Systems Fundamentals; Computer Networks; Database Fundamentals; MySQL; Introduction to Programming; Python; C/OOP fundamentals; Data Structures and Algorithms; Client-Side Technologies; HTML5/CSS3; Web Fundamentals; Bash Shell Script; Cloud and Virtualization Foundations; Software Engineering; Introduction to Generative AI and prompt engineering.
Core Courses	1,203 hrs	67%	Enterprise networking; Linux and Windows administration; Active Directory; virtualization; data center infrastructure; storage and backup; cloud platforms; DevOps and CI/CD; Docker, Kubernetes and OpenShift; Infrastructure as Code; backend and cloud-native development; databases; cybersecurity engineering; penetration testing; incident response; forensics; GRC; cloud security; DevSecOps; AI-assisted operations and security.
Soft Skills	57 hrs	3%	Communication Essentials for Professionals; High Impact Presentations; Job Seeking Skills; Professional Demeanor; Progressive Teamwork; Freelancing basics.
Graduation Project	300 hrs	17%	Capstone project integrating cloud architecture, secure infrastructure, platform automation, cloud-native application deployment, monitoring, incident handling, and documentation.

Integrated core skill map

Networking & Enterprise Infrastructure

- Cisco Network Associate and Internetworking
- Advanced routing and switching design
- Windows Infrastructure and Active Directory
- Red Hat System Administration I, II, III
- Ansible, web server administration, HA clustering

Data Center, Virtualization & Backup

- Information Storage and Management
- Data Center Infrastructure Design
- VMware vSphere and VMware NSX
- OpenStack Deployment, Operation and APIs
- Veeam Backup and Replication
- Nutanix multicloud infrastructure

Cloud Platforms & Cloud Architecture

- AWS SysOps Administration
- AWS Cloud Practitioner/Solutions Architect topics
- Microsoft Azure
- Google Cloud Platform
- Amazon PaaS Web Services
- Cloud security and governance

DevOps, SRE & Platform Engineering

- Docker and Kubernetes
- OpenShift
- Git and version control systems
- DevOps concepts and toolchain
- CI/CD: automation, integration and testing
- Terraform and Infrastructure as Code
- Configuration management and remote state

Application Platform Development

- Python programming
- Node.js
- Backend development
- Web services and APIs
- Microservices architecture
- Message queuing
- Application containers
- Enterprise applications architecture and integration

Database & Data Services

- MySQL Database
- NoSQL Database
- Oracle SQL and PL/SQL
- Oracle DBA I and II
- Database setup for cloud applications

Cybersecurity Engineering

- Security assessment and ethical hacking
- Metasploit Essentials
- Firewalls Technologies
- Web and Mobile Penetration Testing
- Bug hunting
- Malware analysis and reverse engineering
- OT security

Security Operations, IR & Governance

- SOC fundamentals
- Cyber threat hunting and threat intelligence
- Computer and network forensics
- Incident handling and response
- Active Directory pentesting
- Application security
- Risk assessment and compliance
- AI security and DevSecOps

Student deliverable

Every graduate must complete a capstone graduation project and obtain at least one international certification aligned with their specialization path. Recommended certification paths include infrastructure/cloud operations (RHCSA, CKA, AWS-SAA-C03, Azure Administrator, NCP-MCI, HCCDA), platform development (AWS Cloud Practitioner, CKA, AWS-SAA-C03), or cybersecurity (eWPT, eMAPT, eCIR, NSE 4).

Before you apply

Applicants should review the following resources before the technical exam and interview. These prerequisites combine the interview preparation topics repeated across the five source tracks.

AREA	RESOURCE / TOPIC
Networking	Introduction to Computer Networks - MaharaTech course/view.php?id=37
Linux	Linux Essentials - MaharaTech course/view.php?id=2155
Virtualization	VMware Foundation - MaharaTech categoryid=228
Cybersecurity	Introduction to Cybersecurity - MaharaTech
Programming	Programming Fundamentals - CS50; Python Fundamentals - MaharaTech course/view.php?id=43
Software fundamentals	OOP Fundamentals - MaharaTech categoryid=306; Data Structures and Algorithms; Introduction to Software Engineering
Web & Database	Web Fundamentals - MaharaTech course/view.php?id=36; Database Fundamentals - MaharaTech course/view.php?id=740
Cloud	Introduction to Cloud Computing; AWS Cloud Practitioner Essentials - AWS Skill Builder
ITI	ITI Values & Mission - iti.gov.eg/about-us

APPLY AT

ITI.GOV.EG

Applications open each intake period. Check the website for current dates and deadlines.

Source basis: Cloud Architecture, Cyber Security, System Administration, Cloud Architecture detailed deck, and Cloud Platform Development detailed deck.